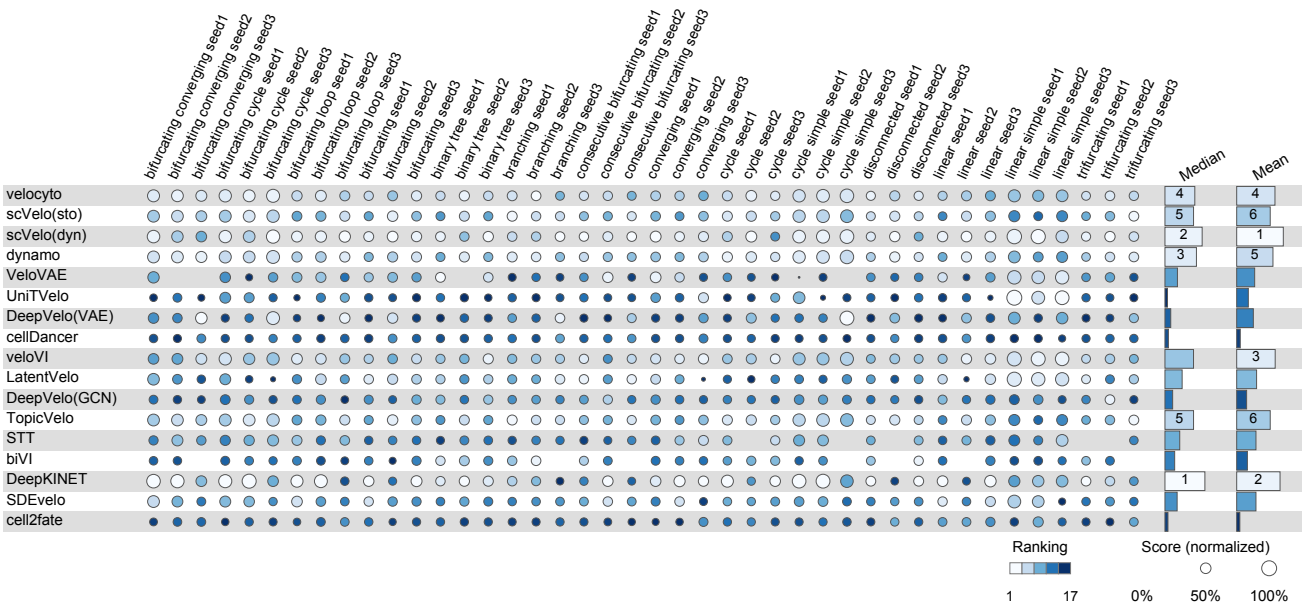
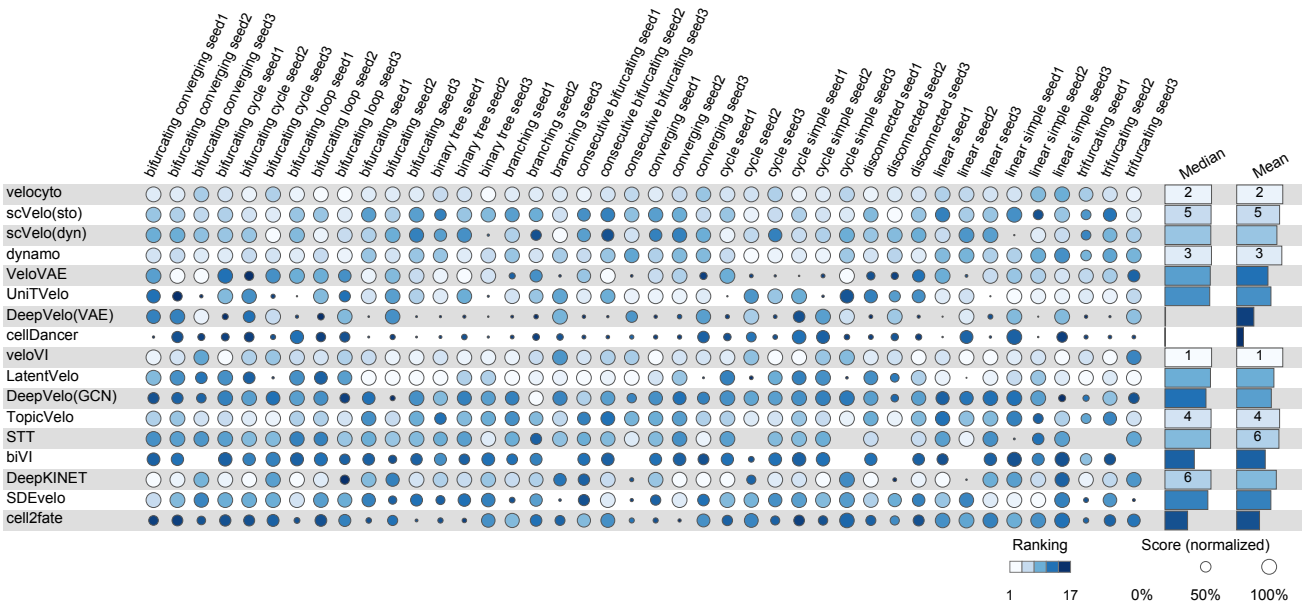
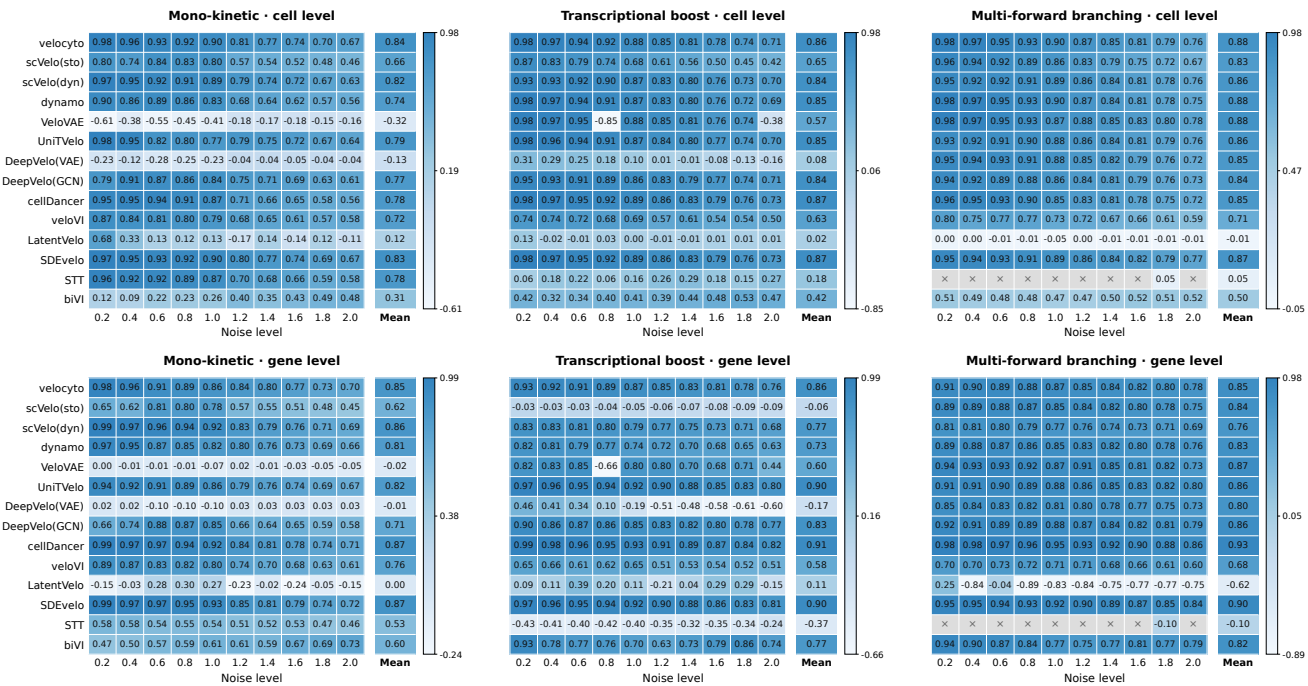
a


Cosine Similarity of High-dim RNA Velocity in Gene Level on Dyngen Simulation Datasets (n=42)
bifurcating converging seed1
bifurcating converging seed2
bifurcating converging seed3
bifurcating cycle seed1
bifurcating cycle seed2
bifurcating cycle seed3
bifurcating loop seed1
bifurcating loop seed2
bifurcating loop seed3
bifurcating seed1
bifurcating seed2
bifurcating seed3
binary tree seed1
binary tree seed2
binary tree seed3
branching seed1
branching seed2
branching seed3
consecutive bifurcating seed1
consecutive bifurcating seed2
consecutive bifurcating seed3
converging seed1
converging seed2
converging seed3
cycle seed1
cycle seed2
cycle seed3
cycle simple seed1
cycle simple seed2
cycle simple seed3
disconnected seed1
disconnected seed2
disconnected seed3
linear seed1
linear seed2
linear seed3
linear simple seed1
linear simple seed2
linear simple seed3
trifurcating seed1
trifurcating seed2
trifurcating seed3
Median
Mean
velocyto 4 4
scVelo(sto) 5 6
scVelo(dyn) 2 1
dynamo 3 5
VeloVAE
UniTVelo
DeepVelo(VAE)
cellDancer
veloVI 3
LatentVelo
DeepVelo(GCN)
TopicVelo 5 6
STT
biVI
DeepKINET 1 2
SDEvelo
cell2fate
Ranking
Score (normalized)
1
17
0%
50%
100%

b


Cosine Similarity of Low-dim RNA Velocity in Cell Level on Dyngen Simulation Datasets (n=42)
bifurcating converging seed1
bifurcating converging seed2
bifurcating converging seed3
bifurcating cycle seed1
bifurcating cycle seed2
bifurcating cycle seed3
bifurcating loop seed1
bifurcating loop seed2
bifurcating loop seed3
bifurcating seed1
bifurcating seed2
bifurcating seed3
binary tree seed1
binary tree seed2
binary tree seed3
branching seed1
branching seed2
branching seed3
consecutive bifurcating seed1
consecutive bifurcating seed2
consecutive bifurcating seed3
converging seed1
converging seed2
converging seed3
cycle seed1
cycle seed2
cycle seed3
cycle simple seed1
cycle simple seed2
cycle simple seed3
disconnected seed1
disconnected seed2
disconnected seed3
linear seed1
linear seed2
linear seed3
linear simple seed1
linear simple seed2
linear simple seed3
trifurcating seed1
trifurcating seed2
trifurcating seed3
Median
Mean
velocyto 2 2
scVelo(sto) 5 5
scVelo(dyn)
dynamo 3 3
VeloVAE
UniTVelo
DeepVelo(VAE)
cellDancer
veloVI 1 1
LatentVelo
DeepVelo(GCN)
TopicVelo 4 4
STT 6
biVI
DeepKINET 6
SDEvelo
cell2fate
Ranking
Score (normalized)
1
17
0%
50%
100%

c


Mono-kinetic · cell level
Transcriptional boost · cell level
Multi-forward branching · cell level
Mono-kinetic · gene level
Transcriptional boost · gene level
Multi-forward branching · gene level
Noise level
Mean